

People Analytics

Groupwork Use Cases 3 – Advanced Turnover Modelling and Business Case Development

Employee turnover is a critical concern for organizations due to its substantial financial and human costs. HR functions frequently report turnover statistics, but descriptive reporting alone may lead to misdiagnosis of problems and poorly targeted interventions.

In this use case, you will apply inferential and predictive analytics to understand voluntary employee turnover at both individual and team levels in a multinational organization. You will then translate analytical insights into a business case for HR action, following the logic and methods presented in Chapter 6: Predicting Employee Turnover.

You will work with two datasets: Individual_Turnover.xlsx and Turnover_Team_Data.xlsx

Prompt 1: Estimating differences in turnover by country

Objective: Explore whether apparent differences in individual voluntary turnover across countries are statistically meaningful.

Tasks:

- Produce a cross-tabulation of Country × Leaver Status.
- Calculate the overall separation rate.
- Conduct a chi-square test of independence to assess whether turnover differs significantly by country. Interpret the results, paying attention to:
 - expected vs observed frequencies,
 - statistical significance,
 - limitations related to small subgroup sizes.

Deliverables:

- Cross-tabulation table with expected frequencies.
- Chi-square test output.
- Interpretation: Should HR be concerned about country differences in individual turnover based on this analysis alone?

Prompt 2: Cross-Country Differences in Team Engagement and Turnover

Objective: examine whether team-level turnover and engagement differ across countries using appropriate inferential tests.

Tasks:

- Generate descriptive statistics (N, Mean, SD, Min, Max) for Team engagement and Team turnover rate by country.
- Conduct a one-way ANOVA for each variable.
 - Test the assumption of homogeneity of variances using Levene's test.
 - Use Welch's ANOVA if variances are unequal.
 - Perform appropriate post-hoc tests to identify which countries differ.

Deliverables:

- Descriptive statistics tables.
- ANOVA output (including Welch correction if required).
- Post-hoc comparison table.
- Interpretation: Which country or countries should HR prioritize for further diagnosis, and why?

Prompt 3: Individual-Level Predictors of Voluntary Turnover

Objective:

Identify individual-level predictors of voluntary turnover.

Tasks:

- Estimate a logistic regression model predicting leaver status using: Gender; Age; Length of service; Performance appraisal rating; Country (as a categorical variable).
- Evaluate overall model fit using: Likelihood ratio test; Nagelkerke R^2 .
- Interpret coefficients using odds ratios.

Deliverables:

- Logistic regression output.
- Table of log-odds and odds ratios.
- Interpretation: Which variables significantly predict turnover? How should HR interpret the effect of gender and performance ratings?

Prompt 4: Gender Differences in Employee Turnover Over Tenure

Objective: Understand when employees are most likely to leave and whether survival patterns differ by gender.

Tasks:

- Conduct a Kaplan–Meier survival analysis using: Length of service as time variable; Leaver status as event indicator; Gender as grouping variable.
- Compare survival curves using a statistical test.
- Visualize survival probabilities over time.

Deliverables:

- Survival table and summary statistics.
- Survival plot.
- Interpretation: At what stages of tenure are gender differences in turnover most pronounced, and what HR processes might this implicate?

Prompt 5 — Team-Level Drivers of Employee Turnover

Objective: Identify team-level drivers of turnover and assess their relative importance.

Tasks:

- Estimate an OLS multiple regression model predicting team turnover using: Engagement; Team leader rating; CSR; Drive for performance; Performance, development and reward; Work-life balance; Country dummy variables (with one reference category).
- Examine: Model fit (R^2 , F-test).
- Unstandardized and standardized coefficients.

Deliverables:

- Regression output.
- Table of standardized coefficients.
- Interpretation: Which team-level factors are most strongly associated with turnover, and how should HR respond?

Prompt 6 — Financial Implications of Gender Differences in Turnover

Objective: Translate analytical findings into financial and strategic implications.

Tasks:

- Estimate the proportion of individual turnover explained by gender alone using logistic regression.
- Use the Nagelkerke R^2 to approximate the share of turnover costs attributable to gender.
- Assume a total annual voluntary turnover cost of £4 million. Calculate potential cost savings if gender-related turnover were eliminated or reduced.

Deliverables:

- Cost calculation.
- Executive-style summary answering: Is there a strong enough business case to justify a targeted retention intervention?
- What cautions should HR keep in mind when interpreting these figures?